



Impact of dietary exposure to polyester microfibers on hematology, serology and histology in a mouse model

Saira Razzaq^a, Safdar Sidra^{a,*}, Arshad Javid^a, Shahid Mehmood^b

^a Department of Wildlife and Ecology, University of Veterinary and Animal Sciences, Lahore, Pakistan

^b Department of Poultry Production, University of Veterinary and Animal Sciences, Lahore, Pakistan

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ABSTRACT

Synthetic fabrics, especially polyester, are a primary source of microplastic fibers (MFs), but there is limited data on their accumulation and dose related health impact in living organisms. This study examined the effects of ingested polyester microfibers (PE-MFs) on hematology, histopathology, and serum biochemistry in albino mice. Mice were given varying doses of PE-MFs (100, 200, 400 and 800 µg/d/mice) for a duration of thirty-five days and a notable decreases in certain hematological parameters such as RBCs, Hb, and platelets, and increases in MCV and MCH was noted at ($p < 0.05$) thereby indicating possible inflammatory response within the body resulting from ingestion of these MFs. Liver enzymes (ALT, AST, and alkaline phosphatase) and histopathological changes in the liver and gastrointestinal tract also exhibited significant variations, with higher levels seen in the group receiving the highest dose of PE-MFs (800 µg/d/mice). In summary, increased exposure to PE-MFs led to a dose-related impact and notable alterations in histopathological, hematological, and serum biomarkers in albino mice. This study highlights the potential hazards associated with dietary exposure to PE-MFs in mammals and emphasizes the need for further research in this field.

1. Introduction

Plastic products have been in use since the 1950s owing to their 'miraculous properties' such as durability, light-weight, low cost, ability to be shaped into different forms and for variety of uses; their non-degradability has converted it into a huge pollution problem worldwide. With an estimated 6.3 billion tons of plastic products manufactured since the last six decades, only about 9 percent of it has ever been recycled (Alabi et al., 2019); hence plastic waste, in particular microplastics (MPs) is considered an emerging environmental contaminant and a growing environmental concern (Napper et al., 2020). Although plastic pollution was already a significant environmental issue prior to the pandemic, the increased use of disposable pandemic-related plastics, such as surgical masks, has further exacerbated the issue (Prata et al., 2020). Despite various legislations on plastics use and recycling efforts in numerous countries, plastic pollution is still prevalent in various environmental settings, including freshwater bodies, oceans, the atmosphere, urban areas, remote locations, and agricultural systems (Rochman, 2018; Wright et al., 2020). Plastic fragmentation into small particles known as microplastics (MPs), defined as having a particle

diameter of less than 5 mm (Thompson et al., 2004), is influenced by factors such as sunlight exposure, mechanical breakdown, and microbial decomposition; however, the specific mechanisms and rates of these processes can vary significantly based on environmental conditions and the types of plastics involved. MPs are listed among the top environmental challenges by the United Nations Environment Program (Derraik, 2002). Additionally, the widespread use of synthetic fabrics, particularly polyester in textiles, is a major source of microplastic fibers (MPFs), a subtype of MPs that pose serious environmental and health risks (Auta et al., 2017). MPFs, once released into the environment, can accumulate in high concentrations, which can vary widely depending on location and environmental conditions, and are readily ingested by a wide range of species, including humans and animals (Zolotova et al., 2022). In fact according to a review by Kwon et al. (2020), almost half of the MPs found in food are mostly microfibers. Upon ingestion, these MPFs can trigger immune responses in the host organisms, leading to cytotoxicity and potential health hazards even in smaller quantities (Von Moos et al., 2012; Mariani et al., 2019; Kwon et al., 2020). These pathophysiological changes in organisms can have serious consequences such as alterations in gene expression, oxidative stress, damage to

* Corresponding author. Dept. of Wildlife and Ecology, UVAS, Pakistan.

E-mail addresses: sairarazzaq702@gmail.com (S. Razzaq), sidra.safdar@uvas.edu.pk (S. Sidra), arshadjavid@uvas.edu.pk (A. Javid), shahid.mehmood@uvas.edu.pk (S. Mehmood).

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epithelial barriers, inflammation and affect the gut micro-flora as well (Harmon, 2018). A detailed and comprehensive review of the impact of MP ingestion or inhalation upon animal health can be found here (Zolotova et al., 2022).

Humans are exposed to MPFs either directly through the atmosphere, sea salt, and drinking water, or indirectly through the food chain (Laitala et al., 2011; van Raamsdonk et al., 2020). Debris from synthetic fabrics, such as polyester in textiles, is considered a major source of exposure to polyester microfibers (PE-MFs) in humans and urban environments (Cox et al., 2019). Additionally, the accumulation of particles in all trophic levels may expose humans to more particles through inhalation or contamination of food sources (Von Moos et al., 2012; Mariani et al., 2019). A recent systematic review (Cesa et al., 2017) of MPFs content in San Francisco Bay (USA) wastewater treatment plants (WWTPs) reported 56 million particles per day ($\geq 125 \mu\text{m}$). Similarly, an Australian WWTP reported 460 million particles per day (ranging from $60 \mu\text{m}$ to $500 \mu\text{m}$). In a German WWTP over a one-year period, it was estimated that microplastics could reach up to 5 trillion particles ($< 500 \mu\text{m}$). When comparing fibers to other shapes of microplastics present in WWTPs, synthetic fibers, mainly composed of polyester, acrylic, polyamide, and polypropylene, were found to be the dominant sources of MPF contamination in WWTPs (Cesa et al., 2017). These studies emphasize the necessity of not only raising public awareness about the continuous intake of plastic into the human body but also implementing improved monitoring and regulatory measures to comprehensively address plastic pollution. However, it remains unclear which types of particles or their sizes may be critical in driving specific health-related conditions and diseases.

"Research on plastic debris, including MPFs, has predominantly focused on marine animals, but it is important to acknowledge that studies have also examined terrestrial animals and humans, revealing alterations in physiology and behavior across various environments (Harmon, 2018; Rezanian et al., 2018). Animal models have also been used to investigate the impacts of MPF on serological indicators, cellular damage, and metabolic changes in gut flora. For example, exposure to plastic-based microfibers led to increased mortality and decreased reproductive efficiency in *Daphnia magna* (Ogonowski et al., 2016). Similarly, microfibers in *Mytilus galloprovincialis* have been found to induce oxidative stress, immune suppression, and cellular apoptosis (D  tre   and Gallardo-Esc  rate, 2017). Moreover, these microfibers are known to absorb toxic substances such as monomers, carcinogenic dyes, pollutants, and pathogens, allowing these contaminants to enter the organisms' bodies and cause toxicity (Wang et al., 2016; Enyoh et al., 2019). Organisms can accumulate plastic waste and associated contaminants through the food chain, posing significant health risks (Nelms et al., 2018). The precise mechanisms through which microplastics and microfibers impact human and animal health are still under investigation; however, emerging evidence suggests that they can lead to inflammation, oxidative stress, and potential disruption of endocrine functions. Hence, it is evident that these minuscule particles have the capability to be harmful, and more research is required to comprehensively grasp their impact on the human body (Novotna et al., 2019; Li et al., 2020). With limited research on the effects of dietary exposure—defined as the consumption of food containing plastic-derived synthetic fibers—particularly in mammalian models, this study aims to assess the impact of polyester microfibers (PE-MFs) on growth performance, hematology, histopathology, and serum biochemistry in albino mice. Polyester was selected owing to its abundance in environmental samples from 60 to 70% reported from multiple studies such as those of De Falco et al. (2019), Ross et al. (2021) and   aravanja et al. (2022). These studies have indicated potential adverse effects of polyester microfibers on health parameters, highlighting the need for further investigation into their implications in dietary contexts. Hence, this study aims to highlight the potential hazards associated with dietary exposure to polyester microfibers in mammals and emphasizes the necessity for additional investigation in this area.

2. Materials and methods

2.1. Experimental site and ethical approval

The present study was conducted at the Department of Wildlife and Ecology, Ravi Campus of the University of Veterinary and Animal Sciences, (UVAS) Lahore. All experimental procedures were carried out in accordance with the Animal Care and Use Committee at UVAS (Approval number: 613/18-10-2022) in alignment with the institution's standard practices for animal research.

2.2. Feed preparation

The standardized mice feed (Feed # 14 consisting of Crude Protein = 14%, Crude Fat = 3%, Crude Fiber = 4%; National Feed Industries Pvt. Ltd.) was ground using a pestle and mortar, with ground feed being batched according to different concentrations of PE-MFs. Each batch was then mixed and manually formed into pellets (approximately 3g each). The experiment utilized polyester fibers sourced from a local pillow manufacturer known for producing high-quality bedding products (relaxsit   Lahore, Pakistan) (Fig. 1a). The obtained fibers were also characterized through FTIR and confirmed to contain the ester functional group (Fig. 1b). These fibers were specifically chosen for the experiment as the manufacturers claimed them to be free from any additives such as flame retardants, antimicrobial agents, or moisture-wicking agents. The decision to use pillow filling polyester fibers as a source of microplastics was made due to its common presence in environmental polyester fiber shedding.

The polyester fibers were cut using a specialized fiber cutter at the manufacturer's site. The fibers were then examined and counted using a Euromex EduBlue digital binocular stereo microscope with a built-in 5.0 MP camera. The average length of the fibers was recorded to be 0.07 mm with an estimated average of 1347 fibers per milligram based on counting under a microscope. Hence 100 μg or 0.1 mg of PE-MFs would approximately contain 134.7 fibers. Based on the study by Messika et al. (2018), different groups were formed according to fiber weight for daily intake per mouse. These groups include the FL-100 group (100 μg = 134.7 fibers approx.), FL-200 group (200 μg = 269.4 fibers approx.), FL-400 group (400 μg = 538.8 fibers approx.), and FL-800 group (800 μg = 1077.6 fibers approx.) (FL = Fiber level). The specified amount of fiber for each group was weighed and mixed manually with the mice's usual feed for the experiment.

2.3. Animals and experimental design

A total of 75 healthy adult male albino mice (*Mus musculus*) with an initial body weight of $28 \pm 5 \text{ g}$ were obtained from the animal breeding house of UVAS. The mice were randomly assigned to five experimental groups with three replicates each (five mice per cage and hence fifteen mice per group) (Fig. 2). The mice were tagged individually by coloring either their tails or ears using marking pens for easy identification. Each group was given one of five dietary levels of PES-MPs in a completely randomized design. The control groups were fed a basal diet, while the other groups were given basal diets supplemented with 100 (FL-100), 200 (FL-200), 400 (FL-400), and 800 (FL-800) $\mu\text{g/d/mice}$ of PE-MFs for 35 days.

2.4. Animal care and management

The mice were kept in sterile cages filled with sawdust bedding, measuring 15 inches in length and 14 inches in width, with 5 mice per cage. The cages were equipped with feeders and water plates. The mice were raised in a controlled environment with temperatures maintained between $25 \pm 2^\circ\text{C}$ and relative humidity between $50 \pm 5\%$. The light and dark cycles in the room were set at 14 h of light and 10 h of darkness. The mice were given food ad libitum, with two feeding times at 6:00 and

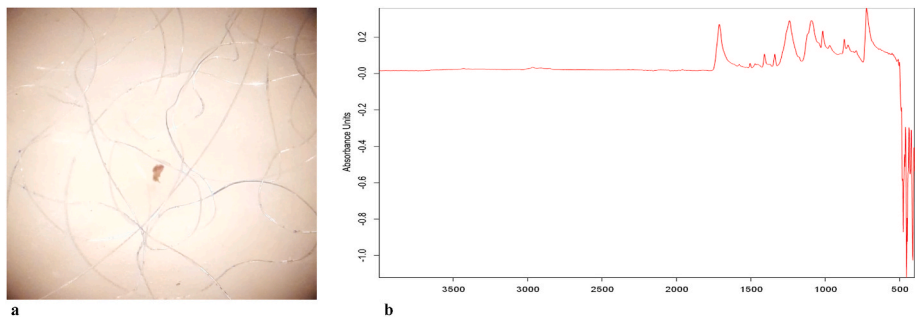


Fig. 1. a: Polyester Microfibers (PE-MFs) were observed under a binocular stereo microscope at 40X magnification, with a scale bar of 100 μm in the image. **Fig. 1b:** Characterization of polyester fibers used for the trial through μFTIR .

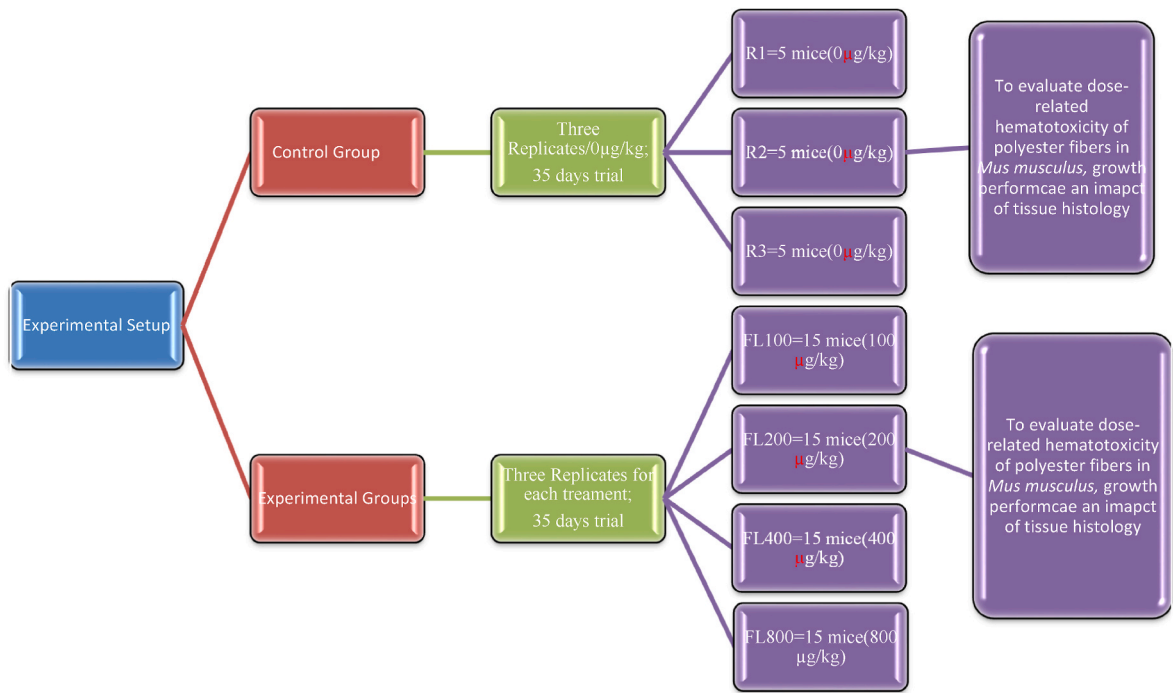


Fig. 2. Experimental setup.

17:00. Fresh water was always available for the mice. The mice were acclimated for seven days before starting a five-week (35 days) feeding trial. Throughout the trial, the mice were monitored daily for their activity levels, water intake, and any signs of fever or mortality.

2.4.1. Fecal matter analysis for percentage retention of PE-MFs

Fecal samples were collected daily in sterile containers to prevent contamination and stored in a cool, dry place until analysis. The pre-labeled fecal samples were oven-dried at 60 °C for 3 h and then ground into a fine powder. The ground samples were transferred to beakers containing a 10% KOH solution at 60 °C for overnight digestion

of organic matter. Following this, the samples were filtered through a series of sieves (500 μm mesh) to remove large particles and organic debris, and subjected to density separation in a 20 ml saline solution (145 g of NaCl dissolved in 400 ml of distilled water). The samples were held overnight at room temperature and then filtered again. The contents were subsequently examined under a stereomicroscope for the identification and quantification of microplastics (MPs). PE-MFs were observed in the residue and then quantified by using a stereomicroscope at 40X. For quantification of percentage (%) retention following formula was used (Cole et al., 2016).

Percentage (%) retention =
$$\frac{\text{No. of MPs given in feed (MP}_i\text{)} - \text{No. of MPs present in fecal material (MP}_f\text{)}}{\text{MP}_i} \times 100$$

2.5. Blood sampling and analysis

After 35 days of the trial, three mice were randomly selected from each treatment group. The samples were obtained after 6 h after the last feeding session. The lateral caudal vein was punctured using a sterile tuberculin syringe, and approximately 2 ml of blood was collected from each mouse. The blood samples were immediately transferred to labeled EDTA-coated tubes for hematological analysis using an automated hematology analyzer (MEK6550) for the analysis of hematological parameters such as red blood cells (RBCs), white blood cells (WBCs), differential WBCs, RBCs, blood platelets, hemoglobin (Hb), hematocrit (HCT), mean corpuscular hemoglobin (MCH), platelet distribution width (PDW), mean corpuscular hemoglobin concentration (MCHC), and mean corpuscular volume (MCV).

The remaining two mice per treatment group had their blood collected and transferred to non-EDTA coated tubes, which were then centrifuged at 3000 rpm for 15 min to harvest serum. The serum was stored at -20°C until further analysis. The liver enzymes such as, alanine transaminase, aspartate transaminase, alkaline phosphatase, and liver and kidney function-associated biomarkers such as bilirubin, total protein, albumin, creatinine, urea, sodium, potassium, and chloride in the serum were analyzed using an automated biochemistry analyzer (RX Daytona plus, Randox Laboratories, United Kingdom).

2.6. Histopathological analysis

After 35 days of PE-MFs exposure, the animals were euthanized by cervical dislocation, dissected, and their isolated organs (liver, stomach, small and large intestine) were washed with normal saline, dried, and preserved in 10% formaldehyde solution in plastic containers for histopathological analysis. The slides were prepared following the methodology outlined by Gartner and Hiatt (2006). In brief, 4- μm thick paraffin-embedded sections of each tissue were cut in duplicate using a semi-automatic microtome (AMOS, AME450, Australia). The sections were then stained with hematoxylin and eosin (H&E), and allowed to dry overnight before examination. Images of the slides were taken at 40X magnification using an optical microscope (LX400, LABOMED, Capelle aan den IJssel, The Netherlands) equipped with a digital camera (DC, 1355-F050, CMEX Euromex, Holland).

2.7. Statistical analysis

The data were analyzed using a one-way ANOVA in SPSS (v.21.0, IBM, Inc.). Mean comparisons were conducted using the Tukey test, with results considered significant at $p < 0.05$.

3. Results

3.1. Effect of PE-MFs particles on blood parameters

There was no potential differences in daily food intake between the control and exposed mice.

Fecal matter was analyzed for presence of PE-MFs. Only a small fraction of the ingested microplastic fibers was found to be excreted hence indicating that the major proportion was retained within the GI tract. The mean number of microplastic fibers retrieved from the fecal samples of each experimental group were as: FL-100 (2.75 ± 1.3 MFs), FL-200 (2.5 ± 2.1 MFs), FL-400 (1.5 ± 1.7 MFs) and FL-800 (5.75 ± 0.9 MFs). This indicates that a major fraction of microplastics is being retained within the GI tract.

Table 1 presents the average levels of hematological biomarkers in albino mice that were fed diets containing different amounts of PE-MFs. The levels of WBCs and PCT were significantly lower ($p < 0.05$) in the group that received the highest dose of PE-MFs (FL-800) compared to the other treatment groups. In contrast, the levels of WBCs and PCT were almost similar ($p > 0.05$) among the control group and the other levels of

Table 1

The impact of 35 Days of exposure to polyester microfibers (PE-MFs) at various dietary levels on hematological biomarkers in albino mice. Data Presented as means $\pm$ SEM.

Blood Biomarkers	Polyester Microfibers (PE-MFs) levels ^a				
	Control	FL-100	FL-200	FL-400	FL-800
WBCs ^b ($10^6/\mu\text{L}$)	21.31 $\pm$ 0.02 ^b	21.42 $\pm$ 0.03 ^b	21.00 $\pm$ 0.28 ^b	22.51 $\pm$ 0.02 ^b	12.95 $\pm$ 1.20 ^a
RBCs ^c ($10^6/\mu\text{L}$)	9.47 $\pm$ 0.07 ^c	9.24 $\pm$ 0.01 ^c	7.62 $\pm$ 0.32 ^b	7.34 $\pm$ 0.12 ^b	6.36 $\pm$ 0.17 ^a
Hb ^d (g/dL)	16.81 $\pm$ 0.01 ^c	16.42 $\pm$ 0.02 ^c	14.60 $\pm$ 0.25 ^b	14.91 $\pm$ 0.05 ^b	10.60 $\pm$ 0.21 ^a
HCT ^e (%)	46.73 $\pm$ 0.042 ^c	45.73 $\pm$ 0.042 ^c	41.1 $\pm$ 1.131 ^b	40.93 $\pm$ 0.049 ^b	30.45 $\pm$ 0.353 ^a
MCV ^f (fL)	43.13 $\pm$ 0.04 ^a	43.99 $\pm$ 0.12 ^a	50.0 $\pm$ 1.83 ^b	50.24 $\pm$ 0.05 ^b	57.50 $\pm$ 0.14 ^c
MCH ^f (pg)	15.61 $\pm$ 0.01 ^a	16.23 $\pm$ 0.32 ^a	18.21 $\pm$ 0.46 ^b	18.11 $\pm$ 0.03 ^b	20.62 $\pm$ 0.434 ^c
MCHC ^g (g/dL)	39.72 $\pm$ 0.03 ^c	38.85 $\pm$ 0.12 ^c	36.41 $\pm$ 0.56 ^b	36.02 $\pm$ 0.05 ^b	28.52 $\pm$ 0.45 ^a
PLT ^h ($10^6/\mu\text{L}$)	403.32 $\pm$ 4.424 ^c	395.53 $\pm$ 5.43 ^c	337.51 $\pm$ 1.81 ^b	338.63 $\pm$ 3.5 ^b	234.54 $\pm$ 4.12 ^a
RDWCV ⁱ (%)	23.92 $\pm$ 1.02 ^c	24.01 $\pm$ 2.32 ^c	18.46 $\pm$ 1.09 ^b	18.31 $\pm$ 1.04 ^b	15.58 $\pm$ 1.02 ^a
RDWSD ^j (fL)	44.13 $\pm$ 0.04 ^c	43.95 $\pm$ 0.91 ^c	40.75 $\pm$ 0.19 ^b	40.72 $\pm$ 0.35 ^b	32.2 $\pm$ 0.08 ^a
PCT ^k (%)	0.50 $\pm$ 0.07 ^b	0.52 $\pm$ 0.32 ^b	0.46 $\pm$ 0.04 ^b	0.54 $\pm$ 0.0 ^b	0.24 $\pm$ 0.06 ^a
MPV ^l (fL)	6.10 $\pm$ 0.07	6.15 $\pm$ 0.24	6.90 $\pm$ 0.01	6.50 $\pm$ 0.00	6.45 $\pm$ 0.49
PDW ^m (%)	13.71 $\pm$ 0.01 ^a	14.09 $\pm$ 0.14 ^a	16.92 $\pm$ 0.70 ^b	16.51 $\pm$ 0.01 ^b	18.52 $\pm$ 0.42 ^c

a-d superscripts within the same row meaning a significant difference ($p < 0.05$), while sharing superscripts among cells of the same row means no statistical difference ($p > 0.05$).

^a Healthy adult male albino mice were divided into five experimental groups, with the control group receiving a basal diet and the other groups receiving various doses (100–800 $\mu\text{g}/\text{d}/\text{mice}$) of PE-MFs.

^b WBC=White blood cells.

^c RBC=Red blood cells.

^d Hb = Hemoglobin.

^e HCT = Hematocrit; 6MCV = .

^f MCH = Met hematocrit.

^g MCHC = Mean corpuscular hemoglobin concentration.

^h PLT = Alanine phosphatase.

ⁱ RDWCV= Alanine transaminase.

^j RDWSD = Aspartate transaminase.

^k PCT = Aspartate transaminase.

^l MPV= Aspartate transaminase and.

^m PDW= Aspartate transaminase.

PE-MFs. Furthermore, the levels of RBCs, Hb, platelets, PDW, RDWCV, RDWSD, and HCT decreased significantly ($p < 0.05$) as the dietary PE-MFs levels increased, with the lowest levels observed in the FL-800 group. On the other hand, the levels of MCV and MCH increased significantly ($p < 0.05$) with higher levels of dietary PE-MFs, specifically in the FL-800 group. However, the level of MPV did not show any significant difference with the varying levels of dietary PE-MFs ($p > 0.05$).

3.2. Effect of PE-MFs particles on serum biochemical indices and liver enzymes

The effects of PE-MFs particles on serum biochemical indices and liver enzymes were examined in albino mice. Table 2 presents the serum biochemical biomarkers and their normal reference values for the mice. The levels of total serum protein, bilirubin, and albumin showed a significant increase ($p < 0.05$) with higher dietary PE-MFs levels, with the highest levels recorded in the FL-800 fed group. Similarly, serum creatinine and urea concentrations were also higher ($p < 0.05$) in the FL-800 group compared to other treatments. The concentrations of serum sodium, chloride, and potassium were significantly higher ($p < 0.05$) in

Table 2
The impact of 35 Days of exposure to polyester microfibers (PE-MFs) at various dietary levels on serum biochemical indices in albino mice. Data Presented as means ± SEM.

Parameters	Ref. Values	Polyester Microfibers (PE-MFs) levels ^a				
		Control	FL-100	FL-200	FL-400	FL-800
Bilirubin (mg/dL)	0.05–0.15	0.13 ± 0.01 ^a	0.12 ± 0.21 ^a	0.26 ± 0.01 ^b	0.27 ± 0.01 ^b	0.36 ± 0.03 ^c
Total serum protein (g/dL)	5.2.0–7.10	5.57 ± 1.01 ^a	5.92 ± 1.24 ^a	7.38 ± 0.97 ^b	8.38 ± 0.11 ^b	9.47 ± 0.12 ^c
Serum albumin (g/dL)	3.4.0–4.80	3.54 ± 0.07 ^a	3.87 ± 0.32 ^a	5.46 ± 0.19 ^b	5.97 ± 0.02 ^b	7.57 ± 0.50 ^c
Albumin/Globulin ratio (A/G ratio)	1.58–2.67	1.80 ± 0.07	1.75 ± 0.07	1.76 ± 0.13	1.77 ± 0.04	1.71 ± 0.07
Serum creatinine (mg/dL)	0.20–0.50	0.21 ± 0.00 ^a	0.25 ± 0.01 ^a	0.45 ± 0.07 ^b	0.62 ± 0.02 ^c	0.73 ± 0.01 ^c
Serum urea (mg/dL)	10.70–20.0	19.4 ± 1.23 ^a	20.4 ± 0.92 ^a	24.6 ± 0.18 ^b	24.4 ± 0.11 ^b	32.8 ± 0.50 ^c
Serum sodium (mmol/L)	142.0–151.0	123.5 ± 0.70 ^a	123.7 ± 0.42 ^a	151.5 ± 1.53 ^b	156.5 ± 0.70 ^c	158.0 ± 0.91 ^c
Serum potassium (mmol/L)	3.82–5.55	3.52 ± 0.45 ^a	3.92 ± 0.82 ^a	7.74 ± 0.15 ^b	6.85 ± 0.49 ^b	10.0 ± 0.00 ^c
Serum chloride (mmol/L)	100.0–106.0	93.5 ± 2.12 ^a	94.25 ± 2.54 ^a	102.5 ± 3.53 ^b	122.5 ± 2.12 ^c	148.5 ± 2.77 ^d

^{a-d} superscripts within the same row meaning a significant difference ($p < 0.05$), while sharing superscripts among cells of the same row means no statistical difference ($p > 0.05$).
^a Healthy adult male albino mice were divided into five experimental groups, with the control group receiving a basal diet and the other groups receiving various doses (100–800 µg/d/mice) of PE-MFs.

mice fed with the highest dose (FL-800) of PE-MFs compared to other treatments. However, the serum albumin to globulin ratio did not show significant differences across treatments.

Table 3 provides the liver function-associated serum enzymes and their normal reference values for the mice. The supplementation of PE-MFs in the diet influenced ($p < 0.05$) the concentrations of alanine transaminase, aspartate transaminase, and alkaline phosphatase in the serum. The levels of these enzymes increased significantly ($p < 0.05$) with higher dietary PE-MFs levels, with the lowest levels recorded in the FL-800 group.

3.3. Histopathology of the liver, stomach, small and large intestine of albino mice exposed to PE-MFs for 35 days

In this study, histopathology was conducted to examine

inflammation and tissue damage in PE-MFs fed groups, as illustrated in Fig. 3. The control group displayed normal integrity and structure of the liver Majority of hepatic cords and hepatocytes are normal. Some focal area show degeneration and necrosis and there is infiltration of mono-nuclear inflammatory cells. and stomach organs. However, there were mild hemosiderin accumulation and karyolysis observed in the small and large intestines The villi are intact. There is no degeneration, sloughing and ulceration in the epithelium. respectively, which could be attributed to aging, dietary factors, or changes in metabolism. On the other hand, mice fed the lowest dose of PE-MFs (FL-100) showed mild liver degeneration and the central vein is normal no cellular changes are seen in the centrilobular hepatocytes. Midzonal and periportal hepatocytes show cellular swelling and degeneration. Neutrophil infiltration in the stomach mucosal layer. The small and large intestines of the FL-100 group appeared normal without significant changes. Subsequent PE-MFs (FL-200) fed groups exhibited karyolysis, lymphocyte infiltration, congestion, and dense epithelium in the liver tissue. In contrast, the FL-400 fed group showed liver mild sinusoidal congestion many hepatocytes show cellular swelling and are full of variable sized vacuoles and no changes are seen in the central vein. Neutrophil infiltration with mild necrosis in the stomach mucosal layer, the gastric epithelium is intact. There is no degeneration, sloughing and ulceration. Mild congestion is seen in the glandular area along with high lymphocyte infiltration in the intestine tissue sections. The liver tissue sections of the highest PE-MFs (FL-800) fed mice displayed hyperplastic hepatocytic regions, cellular degeneration around the central line, and dense lymphocytic infiltration. The stomach tissue sections revealed dense lymphocytic and neutrophil infiltration, as well as hemosiderin deposits, necrosis, and vacuolation. Additionally, there was mild intestinal necrosis and vacuolation in the mice exposed to the highest PE-MFs (FL-800) doses.

4. Discussion

Plastic debris is resistant to biodegradation and generates micro-plastics during environmental abrasions such as mechanical shredding and sunlight effects. The microplastic fibers deposited as granules through ingestion or inhalation could be toxic to a range of physiological processes, including fibrosis and carcinogenesis in mammalian species (Napper et al., 2020). The severity of the toxicity depends on the size, chemical constituents, and level of ingested microfiber waste (Rochman, 2018; Wright et al., 2020). It is well established that microfiber waste decreases the growth performance of living organisms by affecting their digestive system, causing immune toxicity, and disrupting hepato-renal functionality (Nelms et al., 2018; van Raamsdonk et al., 2020). A recent study by Yang et al. (2023) reported the average load of microplastic fibers and fibrils released from multiple wash cycles (500 rubs) of different synthetic garments to be 0.3–18.7 mg/g of fabric for MPFs and 0.0–0.7 mg/g of fabric for fibrils hence an ultimate source of micro-plastic fibers in the environment. These fibers once released in the environment can find their way up the food chain and accumulate within the bodies of exposed living beings.

Growth can be an indicator of normal physiological functioning, the safety of the diet, and the welfare index of the animals. Suppression of

Table 3
The impact of 35 Days of exposure to polyester microfibers (PE-MFs) at various dietary levels on liver enzymes in albino mice. Data Presented as means ± SEM.

Parameters	Reference Values	Polyester Microfibers (PE-MFs) levels ^a				
		Control	FL-100	FL-200	FL-400	FL-800
Alanine transaminase (U/L)	18.0–45.0	34.51 ± 2.12 ^a	41.32 ± 0.32 ^b	53.00 ± 1.41 ^c	54.52 ± 0.70 ^c	65.23 ± 0.02 ^d
Aspartate transaminase (U/L)	74.0–143.0	111.6 ± 1.41 ^d	121.23 ± 0.26 ^b	166.5 ± 4.24 ^c	169.0 ± 2.82 ^c	188.5 ± 0.70 ^d
Alkaline phosphatase (U/L)	62.0–230.0	64.52 ± 3.53 ^d	75.31 ± 1.15 ^b	91.51 ± 2.12 ^c	91.52 ± 0.70 ^c	103.01 ± 1.41 ^d

^{a-d} superscripts within the same row meaning a significant difference ($p < 0.05$), while sharing superscripts among cells of the same row means no statistical difference ($p > 0.05$).
^a Healthy adult male albino mice were divided into five experimental groups, with the control group receiving a basal diet and the other groups receiving various doses (100–800 µg/d/mice) of PE-MFs.

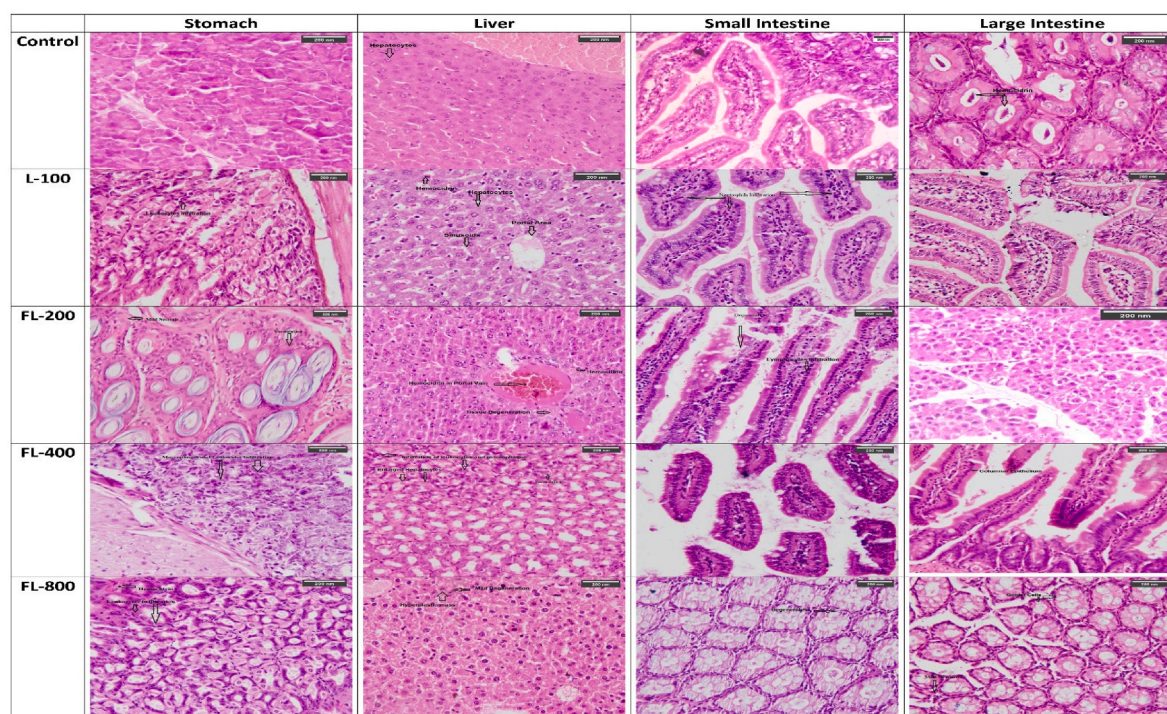


Fig. 3. Photomicrographs of albino mice stomach, liver, small intestine, and large intestine stained with hematoxylin and eosin (H&E). Images were taken using an optical microscope (40X, scale bar 200 μ m). Control group: normal basal diet; FL-100 group: 100 μ g/d/mice of polyester microfibers (PE-MFs) in diet; FL-200 group: 200 μ g/d/mice of PE-MFs in diet for 35 days. FL-400 group: 400 μ g/d/mice of polyester microfibers (PE-MFs) in diet; FL-800 group: 800 μ g/d/mice of PE-MFs in diet for 35 days. Results show normal histological structures in control group, mild to changes in FL-200 and FL-400 group, and more severe effects in FL-800 group.

growth can be due to compromised immune function, intoxication by ingested materials, abnormalities of metabolic processes, and chronic illness (Luo et al., 2021). There are a few studies exploring the toxicity of MPFs upon invertebrate species such as mussels (Walkinshaw et al., 2023), *Daphnia* (Jiang et al., 2023) and different soil invertebrates (Selonen et al., 2020) indicating drastic effects on organism's health. However studies on polyester toxicity in vertebrates is limited to a few studies on aquatic organisms such as fish (Kim et al., 2023). While plastic debris and microplastics have been extensively studied for their effects on marine animals, including fish, altering their physiology and behavior, there is limited data available on the dietary exposure to plastic-derived synthetic fibers such as polyester, particularly in mammalian models and also upon hematology and histology as investigated in the current study. Therefore, this study was designed to evaluate the dietary levels of polyester microfibers and their impact on growth performance, hematology, histopathology, and serum biochemistry in albino mice.

In the current study, different levels of PE-MFs in the diet (ranging from 100 to 800 μ g/d/mice) were found to reduce the growth performance in terms of body weight of albino mice following 35 days of continuous exposure. While no mortality was reported during the trial, our results showed that higher concentrations of PE-MFs led to a significant inhibition of growth in the mice. These results align with the findings of Jin et al. (2021) and Huang et al. (2022), who observed reduced growth performance in mice and fish, respectively, when exposed to various sources of microfibers. The diminished growth performance may be due to the toxic effects of PE-MFs, as plastic-based microfibers are known to cause a range of adverse physiological effects, including disruptions to the digestive system, decreased food intake, and compromised growth (Zolotova et al., 2022). Notably, the exact mechanisms through which microplastics or microfibers affect humans or animals are still not fully understood. However, it is evident that these small particles have the potential to be harmful in the long run with no immediate risk of mortality as indicated by no mortality in our

trial of continuous exposure for thirty-five days, and more research is necessary to fully grasp their impact on the human body (Novotna et al., 2019; Li et al., 2020).

Next, to explore the effect of PE-MFs on the blood system of mice, we first examined hematological parameters, which are well-known biomarkers for detecting pathophysiological changes in living organisms (Mushtaq et al., 2022). Our results showed no significant differences in the levels of WBC and PCT between the control group and the groups that received varying doses of PE-MFs, except for the group that received the highest dose (800 μ g/d/mice), which exhibited a significant decrease. This indicates the potential toxicity of polyester fibers in compromising and suppressing the immune system of mice at higher doses (Guo and Rondina, 2019). However, RBC, Hb, HCT, MCHC, plt, and PCT levels all showed significant decreases with increasing doses of PE-MFs. This change in blood parameters may be due to the interference of toxic particles with the production and function of blood cells, resulting in anemia and decreased oxygen transport to tissues and organs (Stegner et al., 2019). Furthermore, the levels of MCV and MCH were higher compared to the control group, which may indicate macrocytic anemia or in conditions mimicking liver infection. Consistent with our findings, several studies have reported decreases in blood parameters after exposure to toxins or microplastic fibers (Marques et al., 2006; Geens et al., 2010; Hamed et al., 2019). Overall, our findings indicate that exposure to higher doses of PE-MFs particles may adversely affect the blood system by inducing anemia and hepatic injury. These effects potentially contribute to a reduced growth rate. It is well established that the liver is a critical organ in the body, as its impaired health can result in changes in functionality and have drastic effects on animal survival (Mushtaq et al., 2022; Zolotova et al., 2022). Various serum biochemical indices such as serum total protein, globulin, albumin, creatinine, urea, and serum enzymes (ALT, ALP, and AST) are well-known biomarkers used to assess abnormalities in associated tissues (Mushtaq et al., 2022). The concentrations of these metabolites fluctuate when these organs are damaged by toxic substances from

either external (ingested/inhaled) or internal (disease/pathogen) sources. In our experiment, the levels of total serum protein, bilirubin, and albumin in mice increased with higher levels of dietary PE-MFs. Similarly, the activities of liver enzymes ALT, AST, and alkaline phosphatase significantly increased with higher levels of PE-MFs. These results were consistent with the findings of Hamed et al. (2019, 2022), who reported a higher serum biochemical response in fish exposed to microfiber sources. Similarly, changes in serum biochemistry have been observed in *Pomatoschistus microps* exposed to plastic-based polyester, pyrene, and nickel (Oliveira et al., 2013; Norberto, 2014). Thus, changes in biomarker levels in albino mice following exposure to dietary PE-MF particles may result in significant organ damage. To confirm this organ damage post exposure to dietary PE-MFs particles, we conducted histopathology H&E stains of the organs. The results showed damage to the liver, stomach, small and large intestines of the mice, including hepatocyte disorganization, lymphocytic and neutrophil infiltration, necrosis, and vacuolation.

5. Conclusion

This study found that prolonged exposure to increasing amounts of polyester microfibers caused anemia in a mouse model, as evidenced by changes in blood and biochemical markers. The elevated levels of certain serum biomarkers, such as total serum protein, bilirubin, and albumin, along with organ examinations, suggested potential negative effects on blood and liver function. Higher doses of polyester microfibers resulted in more significant changes and tissue damage, particularly in the liver and gastrointestinal tract, indicating potential harm. Further research is needed to understand the underlying mechanisms and assess the possible impact of polyester microfibers on human health.

Ethical approval

All the procedures and protocols used in this experiment were approved by the Animal Use Ethics Committee of the University of Veterinary and Animal Sciences Lahore, Pakistan (NO: 613/18-10-2022).

CRediT authorship contribution statement

Saira Razzaq: Writing – original draft, Methodology, Data curation, Conceptualization. **Safdar Sidra:** Writing – review & editing, Supervision, Methodology, Data curation, Conceptualization. **Arshad Javid:** Supervision, Methodology. **Shahid Mehmood:** Supervision, Data curation.

Consent to participate

Not applicable.

Consent to publish

All authors read and approved the final manuscript.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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